

NAPIER, New Zealand

WMO: 933730

Lat: 39.4703S

Lon: 176.8644E

Elev: 10

StdP: 14.69

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.2	36.3	29.8	23.8	40.2	31.4	25.8	42.0	27.9	54.3	25.2	55.6	3.5	240	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.6	82.9	65.4	79.4	64.4	76.6	63.7	69.1	76.5	67.7	74.5	66.4	72.9	16.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.7	98.3	72.0	65.4	93.7	70.9	63.9	89.0	69.7	33.3	76.9	32.1	74.7	31.1	73.1	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.9	21.8	19.3	DB	29.8	89.5	1.6	2.1	28.6	91.0	27.7	92.3	26.8	93.4	25.7	94.9	(4)
(5)			WB	28.9	71.4	1.6	1.7	27.8	72.6	26.8	73.6	25.9	74.6	24.8	75.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.2	66.0	65.5	62.7	58.2	54.3	50.0	48.9	49.7	52.8	56.1	59.2	63.6	(6)	
	DBStd	7.72	5.28	4.80	4.62	4.73	5.04	4.79	4.36	4.18	4.89	5.13	5.41	5.17	(7)	
	HDD50	240	0	0	0	1	15	59	75	57	25	7	1	0	(8)	
	HDD65	3085	50	46	100	210	333	451	500	474	367	279	187	88	(9)	
	CDD50	2874	496	434	395	248	149	58	41	49	108	196	277	423	(10)	
	CDD65	243	81	60	30	7	2	0	0	0	0	3	14	46	(11)	
	CDH74	1274	475	306	137	15	2	0	0	0	1	13	78	247	(12)	
	CDH80	240	112	59	15	0	0	0	0	0	0	0	6	48	(13)	
(14) Wind	WSAvg	8.5	9.1	8.2	8.1	7.3	7.7	8.0	8.2	8.2	8.7	9.6	9.7	9.2	(14)	
(15) Precipitation	PrecAvg	31.5	1.8	2.3	3.4	2.4	2.7	3.9	3.5	3.1	2.5	1.3	2.0	2.7	(15)	
	PrecMax	35.5	4.3	4.9	9.3	4.2	9.6	11.7	6.2	7.2	5.6	2.6	4.1	6.5	(16)	
	PrecMin	22.1	0.4	0.2	0.2	0.6	0.5	0.7	1.1	0.9	0.9	0.3	0.2	0.2	(17)	
	PrecStd	4.3	1.1	1.6	2.9	1.1	2.2	2.9	1.7	2.1	1.5	0.7	1.1	1.8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.2	86.1	81.9	76.3	72.0	66.7	65.0	65.9	71.6	75.6	80.5	85.1	(19)	
		MCWB	67.1	67.3	64.1	62.6	60.8	55.9	54.8	54.7	57.9	59.4	62.9	65.6	(20)	
	2%	DB	83.3	81.4	77.6	72.2	68.4	63.6	61.7	62.7	67.7	71.8	76.1	80.1	(21)	
		MCWB	65.7	66.3	63.3	61.1	58.2	54.6	52.9	52.9	56.0	58.0	60.5	63.8	(22)	
	5%	DB	79.1	77.7	74.5	69.2	65.9	61.0	59.2	60.1	64.8	68.8	72.7	76.2	(23)	
		MCWB	64.7	65.8	63.4	60.5	57.1	53.1	51.4	51.6	54.4	57.0	59.3	62.7	(24)	
	10%	DB	75.6	74.6	71.9	66.8	63.6	58.9	56.9	57.9	62.0	65.9	69.6	73.2	(25)	
		MCWB	64.2	65.2	62.8	59.8	56.2	52.3	50.6	50.9	53.5	55.9	58.6	62.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.9	71.4	68.4	65.9	62.9	58.5	56.4	56.5	59.4	61.8	65.1	68.7	(27)	
		MCDB	80.0	78.9	74.3	70.2	68.8	62.4	61.5	61.8	67.4	70.4	73.8	78.4	(28)	
	2%	WB	69.1	69.4	66.6	63.9	60.6	56.5	54.5	54.5	57.5	59.9	63.0	66.5	(29)	
		MCDB	77.1	76.0	72.6	67.9	65.4	60.5	59.0	59.1	64.6	67.9	71.0	74.9	(30)	
	5%	WB	67.5	68.1	65.2	62.7	59.0	54.9	52.9	53.2	56.1	58.4	61.6	65.1	(31)	
		MCDB	74.6	74.2	71.1	66.8	63.4	59.1	57.2	57.8	62.0	65.8	69.3	72.5	(32)	
	10%	WB	66.1	66.7	64.0	61.2	57.3	53.2	51.5	52.0	54.7	57.2	60.0	63.8	(33)	
		MCDB	72.7	72.7	69.7	65.8	61.9	57.9	55.8	56.8	60.2	63.7	67.6	70.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	15.6	14.8	15.7	15.2	15.4	14.9	13.6	14.7	15.8	16.1	16.1	15.1	(35)	
		MCDBR	21.7	19.7	20.2	18.0	17.7	17.0	16.2	17.7	19.9	19.9	20.5	20.2	(36)	
	5% WB	MCWBR	10.0	9.2	10.2	10.2	10.6	10.9	10.1	10.8	11.2	10.6	10.8	9.9	(37)	
		MCWBR	16.3	15.5	15.8	13.3	14.3	14.5	13.2	14.6	16.5	16.8	17.5	15.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.340	0.339	0.324	0.312	0.294	0.288	0.289	0.296	0.315	0.326	0.329	0.341	(40)	
		taud	2.559	2.581	2.614	2.631	2.654	2.639	2.642	2.631	2.548	2.531	2.545	2.554	(41)	
	Ebn at Noon		314	307	299	282	269	260	268	284	296	307	316	315	(42)	
		Edh at Noon	34	31	28	24	20	19	20	23	29	33	34	34	(43)	
(44) All-Sky Solar Radiation	RadAvg		2103	1792	1424	996	700	539	602	857	1227	1628	1920	2028	(44)	
	RadStd		128	164	96	71	46	37	47	61	99	99	134	142	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	+0.68	+1.10	N/A	N/A	N/A	-54	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	+0.72	+1.10	N/A	N/A	N/A	-49	-148	N/A	N/A

Nomenclature: See separate page